Citrate

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Identity Sheet

Property	Value
Name	Citrate
Other names	Citric acid (protonated), 2-hydroxypropane-1,2,3-tricarboxylic acid
Chemical formula	$C_6H_8O_7$ (acid); $C_6H_5O_7^{3-}$ (trianion)
Molecular weight	192.12 g/mol
CAS number	77-92-9
PubChem CID	311
pKa values	pKa1=3.13, pKa2=4.76, pKa3=6.40
Solubility	Very soluble (~590 g/L at 20 C for citric acid)

Structural formula:

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HOOC-CH2-C(OH)(COOH)-CH2-COOH
(tricarboxylic acid with central hydroxyl)
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Citrate is a key intermediate in the TCA (Krebs) cycle and the most important **endogenous inhibitor of calcium stone formation** in urine. It chelates calcium, reduces CaOx and CaPO4 supersaturation, and directly inhibits crystal growth and aggregation. Hypocitraturia is a major risk factor for nephrolithiasis.

Molecules Not to Be Confused With

Molecule	Formula	MW (g/mol)	Key Difference
Oxalate	$C_2O_4^{2-}$	88.02	Stone promoter (opposite role); forms CaOx stones
Isocitrate	$C_6H_8O_7$	192.12	TCA cycle isomer; different biological role
Malate	$C_4H_6O_5$	134.09	Another TCA intermediate; dicarboxylic acid

Medical Information

Origin

Endogenous: TCA Cycle Citrate is synthesised in mitochondria by citrate synthase (acetyl-CoA + oxaloacetate -> citrate). It is a central metabolite in the TCA cycle. Plasma citrate: ~100 umol/L. Most urinary citrate derives from plasma filtration and tubular handling.

Exogenous Dietary citrate from citrus fruits (lemons, oranges, grapefruit), berries, tomatoes. Potassium citrate supplements used therapeutically for stone prevention.

Primary & Secondary Biological Roles

Primary role: - **TCA cycle intermediate:** Central to aerobic energy metabolism. - **Urinary stone inhibitor:** Chelates Ca2+ in urine, reducing CaOx and CaPO4 supersaturation.

Secondary roles: - **Fatty acid synthesis precursor:** Cytoplasmic citrate is cleaved by ATP-citrate lyase to oxaloacetate + acetyl-CoA (lipogenesis). - **Acid-base metabolism:** Citrate is metabolised to HCO3-; its excretion represents alkali loss.

Catabolism and Elimination Pathway

- Citrate is freely filtered at glomerulus.
- ~65-90% reabsorbed in proximal tubule via NaDC-1 transporter (sodium-dicarboxylate cotransporter).
- Reabsorption is **increased by acidosis** (reduces urinary citrate – hypocitraturia).
- Reabsorption is **decreased by alkalosis** (increases urinary citrate).
- Remaining 10-35% excreted in urine.

Expression in Humans

Normal Levels

Compartment	Reference Range
Serum citrate	50-130 umol/L
Urinary citrate (24-h)	Male: >250 mg/day; Female: >300 mg/day
Hypocitraturia threshold	Male: <250 mg/day; Female: <300 mg/day

Factors Influencing Levels **Hypocitraturia:** - Metabolic acidosis (most common cause, including distal RTA) - Chronic diarrhoea (alkali loss) - High animal protein diet - Hypokalaemia - UTI (bacteria consume citrate) - Acetazolamide - Idiopathic (up to 40% of stone formers)

Increased citrate: - Alkalosis - Potassium citrate supplementation - High fruit/vegetable diet - Oestrogen (women have higher citrate than men)

Associated Pathologies

Condition	Citrate Pattern	Key Symptoms
Hypocitraturia + nephrolithiasis	<250 mg/day (male)	Recurrent CaOx/CaPO4 stones; ~20-60% of stone formers
Distal RTA	Very low citrate + alkaline urine	CaPO4 stones, nephrocalcinosis, metabolic acidosis
Chronic diarrhoea	Low citrate (metabolic acidosis)	Dehydration, stone risk

Presence in Urine

Should it be normally present? Yes – citrate is a normal and important urinary solute.

Normal urinary levels: >250 mg/day (men); >300 mg/day (women).

Form in urine: At urinary pH 5-7, citrate exists as a mixture of **$Hcit^{2-}$** and **cit^{3-}** anions. These chelate Ca^{2+}, forming soluble calcium-citrate complexes.

Solubility: Citric acid and its sodium/potassium salts are highly soluble. Calcium citrate is moderately soluble (~850 mg/L) – much more so than calcium oxalate.

Detection in Urine

Available Clinical Assays

1. **Citrate lyase enzymatic assay:**
   - **Principle:** Citrate lyase cleaves citrate -> oxaloacetate + acetate. OAA + NADH (MDH) -> malate + NAD+. NADH consumption measured.
   - **Detection:** Absorbance decrease at **340 nm**.
   - **LOD:** ~0.5 mg/dL (~25 umol/L).
   - **Advantages:** Specific, automated (Roche, Beckman).
2. **Ion chromatography:**
   - **Principle:** Anion exchange + conductivity.
   - **LOD:** ~1 umol/L.
3. **Capillary electrophoresis:**
   - **Principle:** Electrophoretic separation + UV at 210 nm.
   - **LOD:** ~5 umol/L.

Optimal Urine Type for Measurement

- **24-hour urine** for stone metabolic workup.
- No special preservatives needed (citrate is stable).
- Refrigerate during collection.
- Report as mg/day or mmol/day.

Actual Gold Standard

Citrate lyase enzymatic assay (NADH at 340 nm) on automated platforms. CV: 3-5%.

Interferences in Measurement

Interference	Effect	Mechanism
Fluorocitrate	Inhibition of citrate lyase	Rare; from fluoroacetate poisoning
Bacterial contamination	Falsely low	Bacteria metabolise citrate
Very low pH	Affects enzyme activity	Assay requires buffered conditions

Research Detection Methods

Spectroscopy Detection (UV-Vis / NIR)

- Citrate has weak UV absorption (<210 nm). Not practical for direct UV detection in urine.
- Enzymatic NADH consumption at 340 nm (see above).
- NIR: Citrate has C-H/C-O overtones but too low concentration for direct NIR.

Fluorescence Detection

- Citrate is non-fluorescent. Indirect:
- **NADH consumption fluorescence:** Decrease in NADH fluorescence (Ex 340 / Em 460 nm) after citrate lyase + MDH. LOD: ~10 umol/L.

- **Eu(III) or Tb(III) complexes:** Citrate enhances lanthanide fluorescence via antenna effect. Ex 278 / Em 545 nm (Tb). LOD: ~1 umol/L.

Raman Detection

Peak (cm-1)	Assignment
~840	C-C stretch
~960	C-OH stretch
~1390	COO- symmetric stretch
~1580	COO- asymmetric stretch

- Conventional Raman: LOD ~10 mM. SERS: LOD ~10-100 umol/L.

FTIR Detection

Band (cm-1)	Assignment	Notes
~1580	COO- asymmetric stretch	Overlaps with urea, proteins
~1390	COO- symmetric stretch	More specific
~1080	C-O stretch	–

- ATR-FTIR: LOD ~5-10 mM. Marginal for urinary concentrations.

Voltammetry Detection

- Citrate is electrochemically inert in the normal potential window.
- **Indirect:** Citrate complexes Cu2+ or Fe3+ -> shifts their redox potentials. LOD: ~10-50 umol/L.
- **Potentiometric membrane electrode:** Citrate-selective ionophore. LOD: ~1 umol/L.

Other Detection Technologies

1. **LC-MS/MS:** LOD ~0.1 umol/L. Definitive.
2. **Biosensor (citrate lyase on electrode):** Amperometric NADH detection. LOD: ~5 umol/L.

Summary Table: Detection Methods Comparison

Method	LOD	Key Parameter	Sample Prep	Strengths
Citrate lyase/NADH	~25 umol/L	340 nm	None	Standard clinical
Ion chromatography	~1 umol/L	Conductivity	Dilution	Multi-anion
Tb(III) fluorescence	~1 umol/L	Ex 278/Em 545 nm	Probe	Sensitive
SERS	~10 umol/L	1390 cm-1	Ag NPs	Label-free
LC-MS/MS	~0.1 umol/L	m/z 191->111	Extraction	Definitive

References

1. StatPearls - Nephrolithiasis. https://www.ncbi.nlm.nih.gov/books/NBK442014/
2. AUA/EAU Guidelines - Metabolic Stone Workup.
3. PubChem - Citric acid, CID 311. https://pubchem.ncbi.nlm.nih.gov/compound/311